

SBOM Manager

Beta Checkpoint Presentation

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Project Overview

Purpose

- Automated SBOM generation across iOS, Android, Windows, macOS, Linux
- Centralized management with web-based interface
- Standards-compliant (SPDX 2.3, CycloneDX 1.4)

Problem Addressed

- Supply chain security vulnerabilities (SolarWinds, Log4Shell)
- Manual SBOM creation is time-consuming and error-prone
- Executive Order 14028 compliance requirements
- Lack of accessible tools for small/medium organizations

Beta Checkpoint Achievements

85% Complete

15

User Stories

18

API Endpoints

75

Source Files

Key Features Delivered in Beta

- Side-by-side Application Comparison with automated difference detection
- Password Reset System with Resend email integration
- Enhanced Dashboard with statistics and quick actions
- Search Autocomplete with real-time suggestions

Three-Tier System Architecture

Frontend (Presentation Tier)

- React + TypeScript with Vite build tool
- 11 pages, 25 components, responsive Tailwind CSS design

Backend (Business Logic Tier)

- FastAPI + Python with Uvicorn ASGI server
- 18 REST API endpoints across 5 route modules
- 4-layer architecture: API / Service / Models / Core

Data Tier & External Services

- PostgreSQL (Supabase) with 5 normalized tables
- Anchore Syft CLI for SBOM generation
- Resend API for email notifications

Feature Spotlight: Application Comparison

Difference Detection

Identifies added, removed, and version-changed components using hash map algorithm ($O(n+m)$ complexity)

Similarity Scoring

Calculates Jaccard index to measure application similarity based on shared components

PDF Export

Generates professional comparison reports with tables, statistics, and visual indicators

Database Architecture

Normalized Schema (4 Tables)

- applications - App metadata with processing status
- components - Deduplicated with UNIQUE(name, version, type)
- application_components - Many-to-many junction table
- contacts - User support requests

Key Design Features

- Component deduplication reduces storage by 70%
- Foreign key constraints ensure referential integrity
- Row-Level Security (RLS) policies for data protection
- Indexed columns for sub-100ms query performance

Alpha vs Beta Progress Comparison

Metric	Alpha	Beta	Change
Total Features	9	15	+66%
User Stories	9	15	+66%
API Endpoints	13	18	+38%
Pages	7	11	+57%
Components	19	25	+32%
Known Issues	10	6	-40%
Completion %	55%	85%	+55%

Quality Metrics & Code Analysis

8,030

Total LOC

6.6%

Comment Ratio

0%

Critical Defects

Known Issues Summary (6 Total)

- 0% Critical Defects (0 issues) - No showstoppers
- 20% Major Defects (3/15) - Syft installation, no vulnerability data, static charts
- 20% Minor Defects (3/15) - Real-time updates, limited search, email dependency
- 40% Reduction in issues from Alpha (10 → 6)

Problems Resolved in Beta

Major Issues Fixed

- Password Reset - Complete email system with Supabase auth integration
- Comparison Feature - Fully functional with PDF export
- Application List UI/UX enhancement – Integrated new filters and pagination
- Search Functionality - Autocomplete with real-time suggestions

Remaining for Final Release

- Vulnerability Integration - NVD API with CVE data (HIGH priority)
- Real-time Updates - WebSocket implementation
- Docker Deployment - Containerization with pre-installed Syft
- Enhanced Search - Full-text search across all components
- Static charts - Metric charts will be displayed

Roadmap to Final Release

Final Release (December 2025)

- Vulnerability database integration with CVSS scoring
- WebSocket for real-time SBOM generation progress
- Docker containerization for simplified deployment
- Automated testing suite (unit, integration, E2E)
- Enhanced search with full-text capabilities

Post-Graduation Vision

- Scheduled vulnerability scanning
- Jira & ServiceNow integration
- Multi-user organizations with RBAC
- Dependency graph visualization
- Mobile applications (iOS/Android)

Thank You!

- 85% Functional Completion - 15/15 user stories in scope completed
- Major Achievement - Full comparison system with PDF export
- Quality Focus - 0% critical defects, ready for limited production
- On Track - Final release targeted for December 2025

Questions?